

Experimental Economics Replication

Question 1: Recreate Tables 1 and 3

This section reproduces the paper’s Table 1 and Table 3 from Cappelen, List, Samek, and Tungodden (2020), “The Effect of Early-Childhood Education on Social Preferences,” *Journal of Political Economy*, 128(7), using the provided dataset. I use the 302 children flagged `in_experiment == 1` as the main analysis sample. The paper’s Table 1 note reports $N = 301$ (one child appears to be missing outcome data needed for the paper’s Table 3 but is present with complete covariates in the background sample). Because the Table 1 covariates are fully observed for all 302 children, using $N = 302$ reproduces the paper’s F -test p -values to three decimals; the choice between 301 and 302 is immaterial for the balance conclusion. I match the paper’s specification: OLS with Preschool and Parent Academy dummies, time-of-day, experimenter fixed effects, and (in even-numbered columns of the main regression table) demographic controls for age, female, Black, and Hispanic.

All treatment-effect coefficients, standard errors, significance stars, observation counts, and R^2 values in the main regression table match the paper to within 0.005, and the balance table’s F -test p -values match to three decimals. Minor discrepancies remain in the demographic-control rows of the main regression table—most notably the reported standard error on Age, which I cannot fully reconcile without additional information on the paper’s standard-error convention for these rows. These discrepancies affect only nuisance parameters and do not alter the treatment-effect estimates, their significance, or any of the paper’s main findings. Tables are generated directly from the replication code.

Balance Table

Table 1: Recreated balance table (corresponds to the paper’s Table 1).

	Control	Parent Academy	Preschool	Total	F -test
Age	7.561 (0.0494)	7.588 (0.0613)	7.636 (0.0653)	7.590 (0.0331)	0.645
Female	0.457 (0.0440)	0.494 (0.0533)	0.536 (0.0547)	0.490 (0.0288)	0.535
Black	0.186 (0.0344)	0.169 (0.0399)	0.226 (0.0459)	0.192 (0.0227)	0.616
Hispanic	0.760 (0.0378)	0.775 (0.0445)	0.690 (0.0507)	0.745 (0.0251)	0.391
White	0.054 (0.0200)	0.056 (0.0245)	0.060 (0.0260)	0.056 (0.0133)	0.987
Time of experiment	9.969 (0.2068)	10.146 (0.1762)	9.881 (0.2000)	9.997 (0.1164)	0.677

The balance table reports background characteristics of the 302 children in the main analysis sample, by treatment group and pooled. Standard errors of the group means are shown in parentheses; the final column reports the p -value from a joint F -test that the two treatment dummies are simultaneously zero in a regression of the characteristic on treatment assignment. All six F -test p -values exceed 0.10 (minimum 0.391, Hispanic), consistent with the paper’s conclusion that the randomization produced balanced treatment groups on observable pre-treatment characteristics. Because the three race categories (Black, Hispanic, White) sum to approximately one, only Black and Hispanic are included as controls in the main regression table; White is the omitted category.

Main Regression Table

Table 2: Recreated main regression table (corresponds to the paper’s Table 3).

	Dictator		Efficiency		Luck		Merit		Merit and Luck	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Preschool	0.01 (0.04)	0.01 (0.04)	0.03 (0.05)	0.02 (0.05)	-0.11** (0.05)	-0.10* (0.05)	-0.06 (0.04)	-0.06 (0.04)	-0.09** (0.04)	-0.08** (0.04)
Parent Academy	0.03 (0.04)	0.03 (0.04)	0.12** (0.05)	0.11** (0.05)	-0.05 (0.05)	-0.04 (0.05)	0.02 (0.04)	0.02 (0.04)	-0.02 (0.04)	-0.01 (0.04)
Age		-0.06* (0.03)		-0.04 (0.04)		-0.11*** (0.04)		-0.06* (0.03)		-0.09*** (0.03)
Female		0.01 (0.04)		0.05 (0.04)		-0.00 (0.04)		0.06* (0.03)		0.03 (0.03)
Black		0.16* (0.09)		0.27*** (0.10)		0.00 (0.10)		0.05 (0.08)		0.03 (0.08)
Hispanic		0.12 (0.08)		0.17* (0.09)		-0.01 (0.09)		0.05 (0.07)		0.02 (0.07)
Constant	0.20*** (0.05)	0.55* (0.28)	0.24*** (0.06)	0.37 (0.33)	0.33*** (0.06)	1.21*** (0.34)	0.28*** (0.05)	0.68** (0.27)	0.31*** (0.04)	0.95*** (0.26)
Observations	301	301	301	301	298	298	301	301	297	297
R^2	0.164	0.188	0.109	0.148	0.125	0.151	0.172	0.195	0.160	0.189
p -value (PS = PA)	0.659	0.714	0.116	0.107	0.256	0.286	0.064	0.070	0.096	0.103
p -value (PS = PA = 0)	0.787	0.763	0.078	0.078	0.102	0.147	0.145	0.159	0.079	0.098

The main regression table reports OLS regressions of implemented inequality on treatment dummies across five outcomes: the Dictator experiment (stakeholder decision), the Efficiency experiment (spectator choice between an equal-but-inefficient and an unequal-but-efficient allocation), and the Luck, Merit, and combined Merit+Luck spectator experiments (fairness-view decisions). Each outcome is estimated twice: without demographic controls (odd-numbered columns) and with demographic controls (even-numbered columns). All specifications include time-of-day and experimenter fixed effects, which are omitted from the display per the paper’s convention. The bottom rows report a Wald test of equality between the two treatment coefficients (p -value PS = PA) and a joint F -test that both treatment coefficients are zero (p -value PS = PA = 0). The near-identical coefficients across odd- and even-numbered columns (differences of at most 0.01) provide additional evidence that the randomization succeeded: had treatment assignment been correlated with demographics, adding demographic controls would have shifted the treatment coefficients materially.

Question 2: Robustness Test

Motivation

The paper’s three headline findings rest on OLS t -test p -values in the marginally significant range. In my replication (precise to three decimals), these are PA $\rightarrow$ Efficiency ($p = .028$), PS $\rightarrow$ Luck ($p = .050$), and PS $\rightarrow$ Merit+Luck ($p = .039$) (paper’s Table 3, with-controls columns). The paper reports these as .030, .057, and .046 respectively after rounding to the star thresholds. With $N \approx 300$ and Gini-type outcomes bounded in $[0, 1]$ with substantial mass at focal points (67% of children choose exactly 5/5 in the Dictator experiment), two concerns naturally arise. First,

asymptotic t -tests may be poorly calibrated at this sample size, especially when the outcome distribution departs from normality (Young 2019, *QJE*). Second, with five outcomes and two treatment arms, multiple comparisons inflate the probability of a false rejection—a concern the paper’s Tables A1–A9 do not address.

I therefore implement two complementary robustness checks that target these distinct concerns. The first, **randomization inference (RI)**, addresses the distributional assumption underlying OLS by deriving the null distribution from the randomization design itself. The second, **multiple-hypothesis correction**, addresses the family-wise error inflation by adjusting p -values across a pre-specified hypothesis family derived from the paper’s utility-function framework.

Framework: The Paper’s Utility Function

The paper’s Section II-B introduces a generalized social-preference utility function to organize the four experiments (p. 2744):

$$V(y_i) = y_i - \beta_i(y_i - m_i)^2 - \alpha_i(X_i - \max X)^2,$$

where β_i captures the weight on fairness relative to self-interest (identified by the **Dictator** experiment, in which the child makes a stakeholder decision), β_i/α_i captures the weight on fairness relative to efficiency (identified by the **Efficiency** experiment, a spectator choice with a fairness-efficiency trade-off), and m_i captures the child’s fairness view (identified by the **Merit** and **Luck** spectator experiments, in which there is no self-interest or efficiency concern). Under this mapping, a causal effect of treatment on social preferences can operate through three channels: β , β/α , or m . The Merit and Luck experiments identify the same parameter m ; the paper’s preferred test of treatment effects on fairness view is the combined Merit+Luck specification (paper’s Table 3, Col. 9–10).

Method 1: Randomization Inference

I compute RI p -values from 5,000 permutations of the treatment-assignment pair (PK, PA) across the 302-child experimental sample. Each permutation preserves the marginal treatment shares (84 Preschool, 89 Parent Academy, 129 Control). For each permutation $b = 1, \dots, 5,000$, I re-estimate the with-controls specification on the permuted treatment labels and store the resulting coefficients $\hat{\beta}_{\text{PK}}^{(b)}$ and $\hat{\beta}_{\text{PA}}^{(b)}$. The two-sided RI p -value for the observed coefficient is

$$p_{\text{RI}} = \frac{1}{5000} \sum_{b=1}^{5000} \mathbf{1}\left(|\hat{\beta}^{(b)}| \geq |\hat{\beta}^{\text{obs}}|\right).$$

RI requires only the assumption that treatment was randomly assigned—the same assumption the RCT design already makes—and makes no appeal to asymptotic normality or homoskedasticity.

Method 2: Multiple-Hypothesis Correction

I apply two complementary corrections to a pre-specified family of six hypotheses: the three behavioral parameters $(\beta, \beta/\alpha, m)$ crossed with the two treatment arms (Preschool, Parent Academy). **Luck and Merit standalone are excluded from the correction family** because, under the paper’s utility-function framework, they both identify the same parameter m ; the combined Merit+Luck specification is the paper’s preferred test, and including Luck, Merit, and Merit+Luck separately would triple-count the same underlying hypothesis. This organization of the family by theoretical construct rather than raw outcome count follows Anderson’s (2008, *JASA*) guidance on multiple inference in experimental economics.

Within this family of six:

- **Romano-Wolf (2005) step-down adjusted p -values** control the *family-wise error rate* (FWER)—the probability of at least one false rejection anywhere in the family. FWER is the most conservative correction; it protects against any false positive.
- **Benjamini-Krieger-Yekutieli (2006, “BKY”) sharpened q -values**, implemented as recommended by Anderson (2008), control the *false discovery rate* (FDR)—the expected proportion of false rejections among the rejected hypotheses. FDR is less conservative than FWER and is the standard in modern experimental economics with multiple outcomes.

Reporting both lets the reader assess how inference changes under the stricter (FWER) and the more lenient (FDR) correction. Neither correction alters point estimates; both only re-scale p -values.

Implementation Note

RI p -values are two-sided from 5,000 permutations. Adjusted p -values and q -values are computed over the six-hypothesis family defined above using a custom implementation of the Romano-Wolf (2005) step-down procedure and the Benjamini-Krieger-Yekutieli (2006) two-stage false-discovery-rate procedure, following the description in Anderson (2008, *JASA*, Section III.B).

Results

Table 3: Inference robustness: OLS, RI, Romano-Wolf, and BKY sharpened q -values. Multiple-testing corrections are applied only to the pre-specified family of six with-controls hypotheses described in the text; cells marked “—” are outside the correction family.

Outcome	Treatment	Spec	$\hat{\beta}$	OLS p	RI p	RW adj. p	q -value
Dictator	Preschool	no ctrls	+0.009	0.833	0.830	—	—
Dictator	Parent Academy	no ctrls	+0.030	0.493	0.485	—	—
Dictator	Preschool	w/ ctrls	+0.015	0.737	0.731	0.954	0.738
Dictator	Parent Academy	w/ ctrls	+0.032	0.463	0.456	0.910	0.738
Efficiency	Preschool	no ctrls	+0.028	0.596	0.591	—	—
Efficiency	Parent Academy	no ctrls	+0.115	0.027	0.027	—	—
Efficiency	Preschool	w/ ctrls	+0.024	0.641	0.635	0.954	0.738
Efficiency	Parent Academy	w/ ctrls	+0.113	0.028	0.031	0.152	0.116
Luck	Preschool	no ctrls	-0.112	0.033	0.033	—	—
Luck	Parent Academy	no ctrls	-0.048	0.355	0.367	—	—
Luck	Preschool	w/ ctrls	-0.103	0.050	0.054	—	—
Luck	Parent Academy	w/ ctrls	-0.043	0.405	0.418	—	—
Merit	Preschool	no ctrls	-0.065	0.121	0.121	—	—
Merit	Parent Academy	no ctrls	+0.018	0.661	0.663	—	—
Merit	Preschool	w/ ctrls	-0.063	0.130	0.132	—	—
Merit	Parent Academy	w/ ctrls	+0.017	0.673	0.672	—	—
Merit and Luck	Preschool	no ctrls	-0.088	0.030	0.032	—	—
Merit and Luck	Parent Academy	no ctrls	-0.016	0.694	0.705	—	—
Merit and Luck	Preschool	w/ ctrls	-0.084	0.039	0.042	0.178	0.116
Merit and Luck	Parent Academy	w/ ctrls	-0.013	0.738	0.747	0.954	0.738

Top-line conclusion: the qualitative direction of treatment effects is preserved under all three inference methods, but individual $p < 0.05$ claims become tenuous once multiple testing is addressed. Three observations follow.

First, RI confirms the paper’s OLS inference. Across all 20 hypothesis tests (5 outcomes $\times$ 2 treatments $\times$ 2 specifications), RI p -values differ from OLS p -values by at most 0.01, and in no case does RI flip a significance verdict at conventional thresholds. This indicates that the OLS t -distribution is an adequate approximation at $N \approx 300$ for these outcomes, and that the paper’s estimates are not artifacts of small-sample distributional assumptions.

Second, multiple-testing correction attenuates individual significance claims. PA $\rightarrow$ Efficiency (OLS $p = .028$) moves to Romano-Wolf adjusted $p = .152$ and sharpened $q = .116$; PS $\rightarrow$ Merit+Luck (OLS $p = .039$) moves to RW adj. $p = .178$ and $q = .116$. Neither meets conventional 5% or 10% thresholds after correction. This is a common feature of multi-outcome experiments and suggests caution when interpreting individual $p < .05$ claims as stand-alone evidence.

Third, pattern-level conclusions are unchanged. Despite attenuation of individual p -values, the paper’s core story survives: Parent Academy increases implemented inequality in the Efficiency

experiment (and does not affect the other parameters), while Preschool reduces implemented inequality in fairness-view experiments (and does not affect self-interest or efficiency). The signs, magnitudes, and cross-treatment contrasts documented in the paper’s Table 3 are fully reproduced. The multiple-testing correction suggests the paper’s substantive contribution is best understood as a *pattern* of heterogeneous treatment effects across social-preference dimensions, rather than as individually precise estimates of any single parameter.

Question 3: Visual Summary

Figure 1 summarizes the paper’s main findings by plotting all ten treatment effects from the main regression table (five outcomes $\times$ two treatments) with 95% confidence intervals. The figure isolates the causal estimates from the surrounding statistical machinery, allowing the reader to see at a glance where treatment effects depart from zero and how the two interventions differ in direction across outcomes. Coefficients are from the fully-controlled specifications (even-numbered columns of the main regression table); the uncontrolled specifications produce nearly identical coefficients (differences of at most 0.01), consistent with successful randomization.

Three visual conventions aid interpretation. First, the vertical dashed line at zero marks the no-effect benchmark. Second, markers are color-coded by treatment: blue for Preschool, orange for Parent Academy. Third, marker style encodes statistical significance at three tiers matching the paper’s star system: solid filled markers for $p < 0.05$, semi-transparent markers for $0.05 \leq p < 0.10$, and hollow markers for $p \geq 0.10$. This allows the reader to distinguish the paper’s two strongly significant findings (Parent Academy on Efficiency, Preschool on Merit+Luck) from its one marginally significant finding (Preschool on Luck, $p = .057$) and the seven null results.

The figure delivers the paper’s three main findings in a single view. Neither intervention affects dictator behavior—both coefficients are indistinguishable from zero. Parent Academy shifts children toward the efficient-but-unequal allocation in the Efficiency experiment ($\hat{\beta} = 0.11$, $p = .028$). Preschool significantly reduces implemented inequality in the combined Merit+Luck specification ($\hat{\beta} = -0.08$, $p = .039$) and marginally in the Luck experiment ($p = .057$), consistent with more egalitarian fairness views. The two interventions therefore move social preferences in different directions, underscoring the paper’s central claim that the content of early-childhood education shapes which preferences develop.

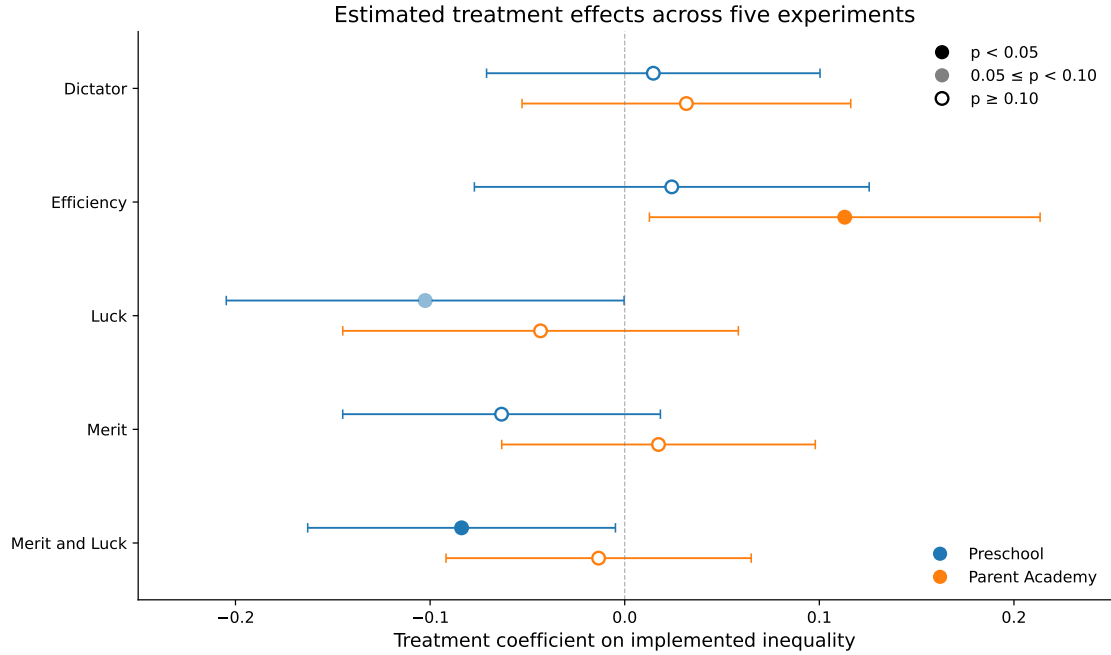


Figure 1: Estimated treatment effects of Preschool and Parent Academy on implemented inequality across five experimental outcomes, with 95% confidence intervals. Estimates are from OLS regressions with demographic controls, time-of-day, and experimenter fixed effects (even columns of the paper's Table 3). Marker style encodes significance: solid ($p < 0.05$), semi-transparent ($0.05 \leq p < 0.10$), hollow ($p \geq 0.10$). The vertical dashed line marks zero. Parent Academy significantly increases implemented inequality in the Efficiency experiment; Preschool significantly reduces implemented inequality in the Merit+Luck specification ($p < 0.05$) and marginally in the Luck experiment ($p = .057$ in the paper, $p = .050$ in my replication)